

Multiple Choice (10 marks)

1. Life on Earth originated from
 - (a) We don't know for sure.
 - (b) aliens (panspermia).
 - (c) comets.
 - (d) the Earth's oceans.
2. What is true about the rotation of a solid disk?
 - (a) The velocity is the same on every point on the disk
 - (b) The part closer to the axis has a longer period
 - (c) The period of rotation is the same on every point on the disk
 - (d) The part closer to the axis has a shorter period
3. A comet of mass m impacts the earth (mass M radius R) at the minimum impact speed. What is the expression for the total energy released in the impact?
 - (a) $m \cdot v$
 - (b) $0.5 \cdot m / (R^3)$
 - (c) $0.5 \cdot m \cdot (2 GM/R)$
 - (d) $0.6 \cdot G(M^2)/R$
4. The distance between the Earth and the star Altair is one million times greater than the distance between the Earth and the Sun. How far is Altair from the Earth?
 - (a) 9.3×10^{13} meters
 - (b) 9.3×10^{10} meters
 - (c) 1.5×10^{14} meters
 - (d) 1.5×10^{17} meters
5. What do meteorites reveal about the solar system?
 - (a) They reveal that the early solar system consisted mostly of hydrogen and helium gas.
 - (b) They reveal that meteorites are much older than the comets and planets.
 - (c) They reveal that the age of the solar system is approximately 4.6 billion years.
 - (d) They reveal that the solar system once contained 10 planets.

True / False (10 marks)

Answer True or False

6. True or False: The Sun's corona is not visible because it emits primarily in the ultra-violet spectrum, which is outside the visible light range.
7. True or False: All forms of life require a source of organic molecules, making it essential for survival and metabolism.
8. True or False: Jupiter's magnetic field is about 20,000 times stronger than Earth's magnetic field.
9. True or False: Type-Ia supernovae occur in binary systems where one star is a white dwarf that accretes material from a companion star.
10. True or False: Earth is the only planet in the solar system that currently exhibits active plate tectonics.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the methods by which scientists determine that most meteorites originate from the asteroid belt, focusing on spectral analysis and its implications for our understanding of asteroid composition.
12. Discuss the significance of the findings from the Martian meteorite ALH 84001, specifically addressing the role of amino acids and the implications of their chirality in the context of extraterrestrial life evidence.

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